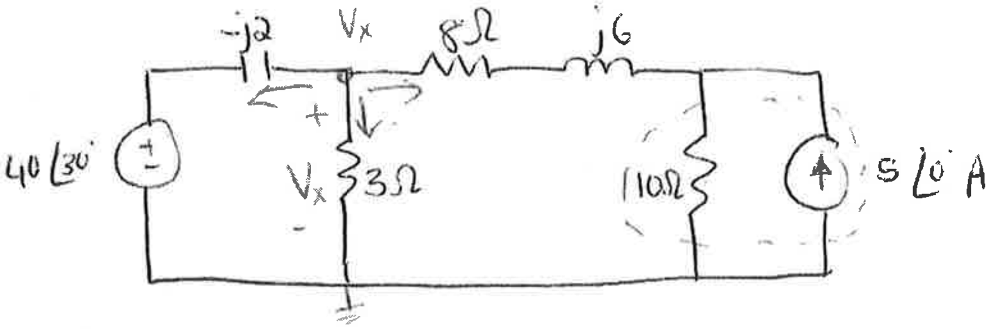
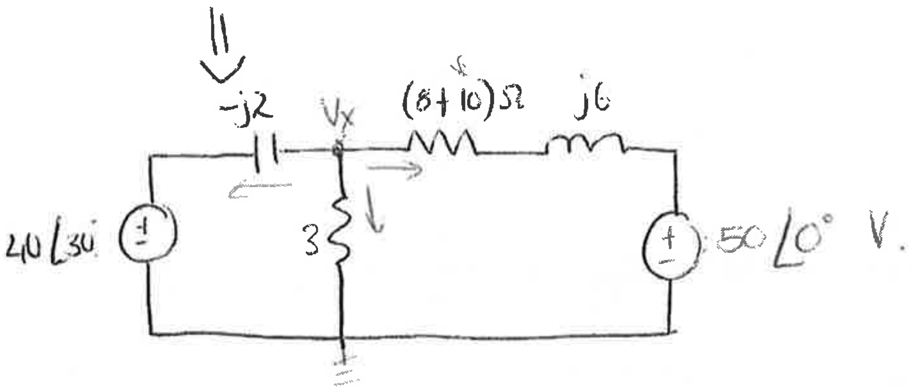
# Tutorial 9

## 10.13




Determine $V_x$.




* Nodal analysis:

$$\frac{V_x - 40\angle 30^\circ}{-j2} + \frac{V_x}{3} + \frac{V_x - 50\angle 0^\circ}{18 + j6} = 0$$

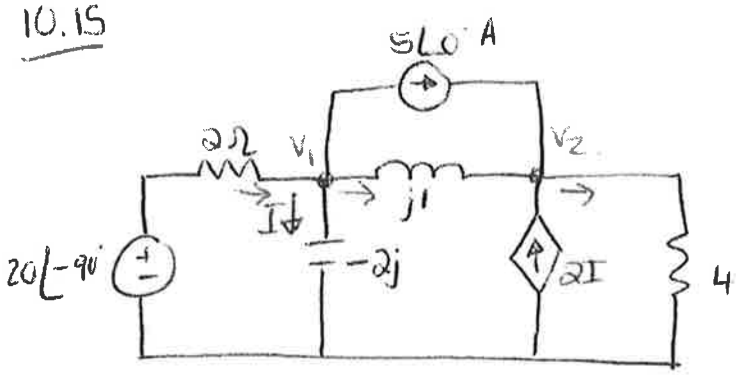
$$j0{,}5\,(V_x - 40\angle 30^\circ) + 0{,}33\,V_x + \frac{V_x}{18{,}97\angle 18{,}4^\circ} - 2{,}635\angle -18{,}4^\circ = 0$$

$$j0{,}5V_x + 0{,}33\,V_x + (0{,}05 - j0{,}0166)\,V_x = 2{,}5 - j0{,}8317 - 10 + 17{,}32j$$

$$(0{,}38 + j0{,}4834)\,V_x = -7{,}5 + j16{,}4883$$

$$V_x = 29{,}4\angle 62{,}8^\circ$$

10.15




Solve for I using nodal analysis

@ Node 1 :

$$\frac{20\angle{-90} - V_1}{2} = 5\angle 0^\circ + \frac{V_1}{-2j} + \frac{V_1 - V_2}{j1}$$

$$\frac{-20j - V_1}{2} = 5 + \frac{V_1}{-2j} + \frac{V_1 - V_2}{1j}$$

$$-10j - 0{,}5V_1 = 5 + j0{,}5V_1 - jV_1 + jV_2$$

$$-5 - 10j = (0{,}5 - 0{,}5j)V_1 + jV_2 \quad \text{——— (1)}$$

@ Node 2 :

$$\frac{V_1 - V_2}{j1} + 5\angle 0^\circ + 2I = \frac{V_2}{4} \qquad * \boxed{I = \frac{V_1}{-2j}}$$

$$-jV_1 + jV_2 + 5 + jV_1 = 0{,}25V_2.$$

$$V_2 = \frac{5}{0{,}25 - j} \quad \text{——— (2)}$$

⇒ (2) in (1)

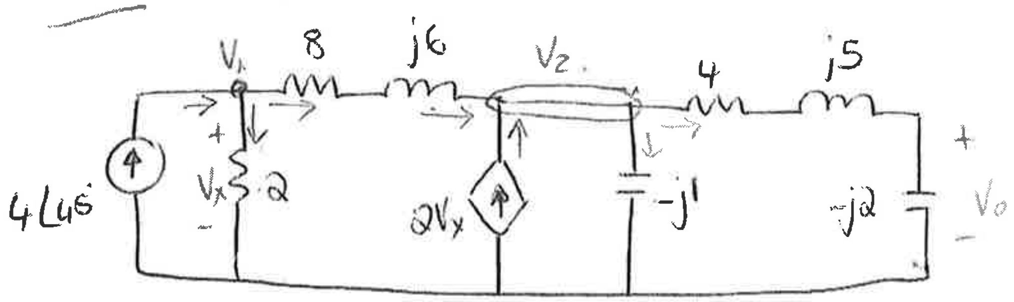
$$-5 - j10 = 0{,}5(1-j)V_1 + j\left(\frac{5}{0{,}25-j}\right)$$

$$(1-j)V_1 = -10 - j20 - \left(\frac{j40}{1 - j4}\right)$$

$$V_1 = 15{,}81\angle{-46{,}5^\circ}$$

$$I = \frac{V_1}{-2j} = 7{,}906\angle 43{,}49^\circ$$

10.18 *




Use nodal analysis to obtain $V_0$.

* ①

$$4\angle 45^\circ = \frac{V_1}{2} + \frac{V_1 - V_2}{8+j6}$$

$$200\angle 45^\circ = (29 - j3)V_1 - (4-j3)V_2 \quad \text{——— ①}$$

* ②

$$\frac{V_1 - V_2}{8+j6} + 2V_x = \frac{V_2}{-j} + \frac{V_2}{4+j5-j2}$$

$$V_1 = \left(\frac{12+j41}{104-j3}\right)V_2 \quad \text{——— ②}$$

* $\boxed{V_x = V_1}$

Substitute ② in ①

$$\Rightarrow 200\angle 45^\circ = (29-j3)\left(\frac{12+j41}{104-j3}\right)V_2 - (4-j3)V_2$$

$$\therefore V_2 = \frac{200\angle 45^\circ}{14.21\angle 89.17^\circ} = 14.07\angle -44.17^\circ$$

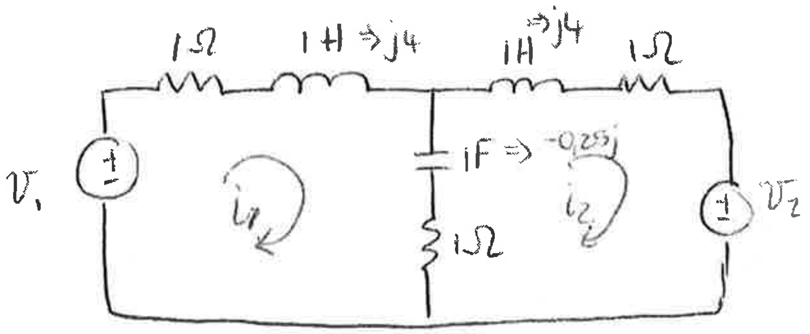
$$\therefore V_0 = \left[\frac{-j2}{4+j5-j2}\right]V_2$$

$$= 0{,}4\angle -126.87\; V_2$$

$$= 5.63\angle -171.03^\circ \Rightarrow 5{,}63\angle 360 + -171{,}03$$

$$= 5{,}63\angle 188{,}97^\circ$$

10.28




Determine $i_1$ and $i_2$.

Let $v_1 = 10\cos 4t$ V

$v_2 = 20\cos(4t - 30°)$ V.

* $1\text{ H} \Rightarrow j\omega L = j4$ $\qquad 1\text{F} \Rightarrow \frac{1}{j\omega C} = -0,25j$
* $V_1 = 10\angle 0°$ $\qquad$ * $V_2 = 20\angle -30°$

Mesh 1:

$$-10 + (2 + 3,75j)I_1 - (1 - 0,25j)I_2 = 0$$

$$(2 + 3,75j)I_1 = 10 + (1 - 0,25j)I_2 \quad \text{—} (1)$$

Mesh 2: $-(1 - 0,25j)I_1 + (2 + 3,75j)I_2 = -20\angle -30°$ — (2)

$$\Rightarrow \begin{bmatrix} (2 + 3,75j) & (-1 + 0,25j) \\ (-1 + 0,25j) & (2 + 3,75j) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 10 \\ -17,32 + j10 \end{bmatrix}$$

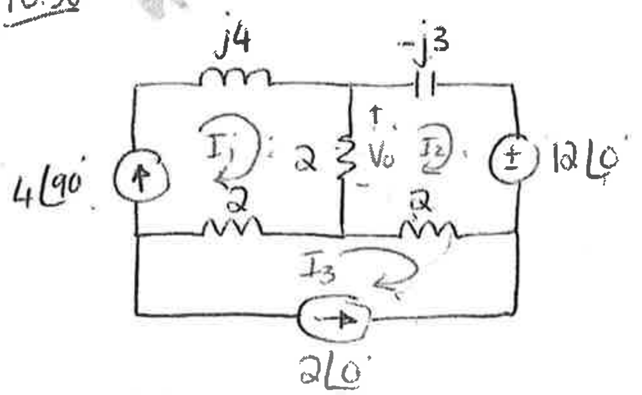
$$\Rightarrow I_1 = 2,741 \angle -41,07°$$
$$I_2 = 4,114 \angle 92°$$

$\therefore i_1(t) = 2,741 \cos(4t - 41,07°)$ A

$i_2(t) = 4,114 \cos(4t + 92°)$ A.

10.36

Compute $V_o$ using Mesh analysis.




$\rightarrow I_1 = 4\angle 90°$
$= 4j$

$\rightarrow I_3 = -2$

* From mesh 2 :

$$-2I_1 + (4 - j3)I_2 - 2I_3 + 12 = 0$$

$$(4 - j3)I_2 = 8j - 2 - 12$$

$$\cancel{I_2 = \frac{-24 + j8}{4 - j3}}$$

$$I_2 = \frac{-16 + j8}{4 - j3}$$

$$= -3{,}52 - j0{,}64$$

$$\therefore V_o = 2(I_1 - I_2)$$

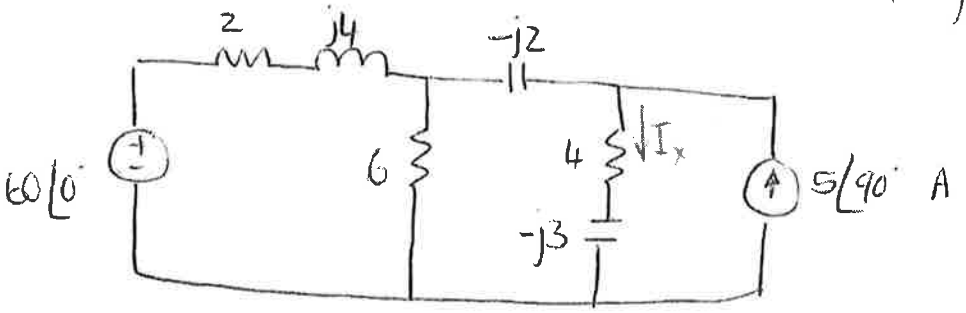
$$= 2(j4 + 3{,}52 + j0{,}64)$$

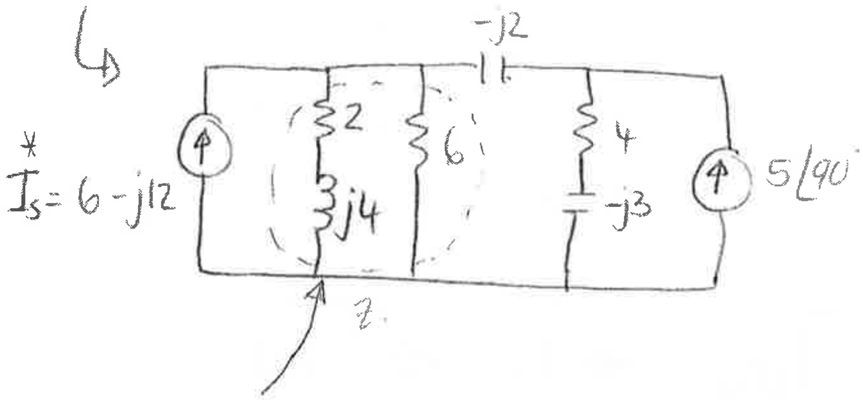
$$= 7{,}04 + j9{,}28$$

$$= 11{,}648\angle 52{,}82° \text{ V}$$

10.52 *

Use source transformation to find $I_x$



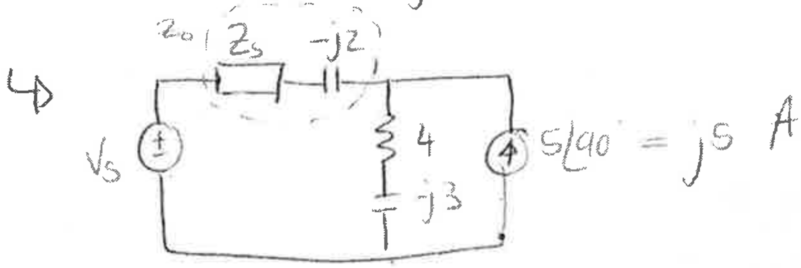




$* \; I_s = \frac{60\angle 0°}{2+j4} = (6 - j12)$ A

$\therefore Z = 6 \| (2+j4) = 2{,}4 + j1{,}8$

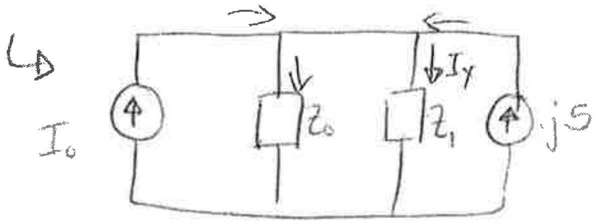
$V_s = I_s Z = (6-j12)(2{,}4+j1{,}8)$

$= 36 - j18$




$\therefore Z_0 = Z_s - j2 = 2{,}4 - j0{,}2$

$\therefore I_0 = \frac{V_s}{Z_0} = 15{,}517 - j6{,}207$



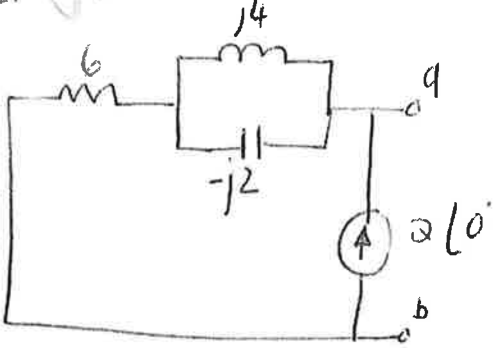

$I_x = \frac{Z_0}{Z_0 + Z_1}(I_0 + j5)$

$= 5 + j1{,}563$

$= (5{,}238 \angle 17{,}35°)$ A

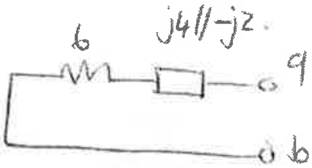
10.56 — Thevenin & Norton.

a)

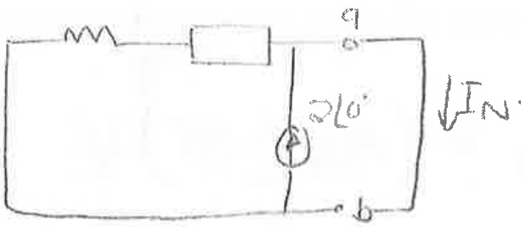



(*) $Z_N = Z_{TH} = 6 + (j4 \parallel -j2)$

$= (7{,}211 \angle -33{,}69^\circ)\ \Omega.$



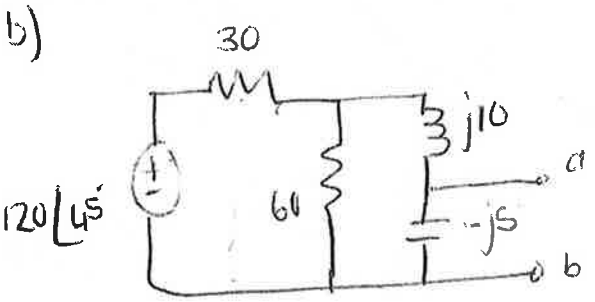

(*)



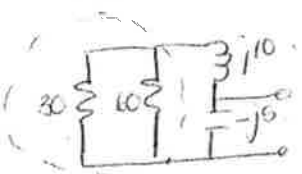

$\Rightarrow I_N = 2\angle 0^\circ$ A.

$\therefore V_{TH} = I_N(Z_{TH}) = (14.422 \angle -33{,}69^\circ)$ V

b)




30//60 = 20.




$* Z_{TH} = Z_N$

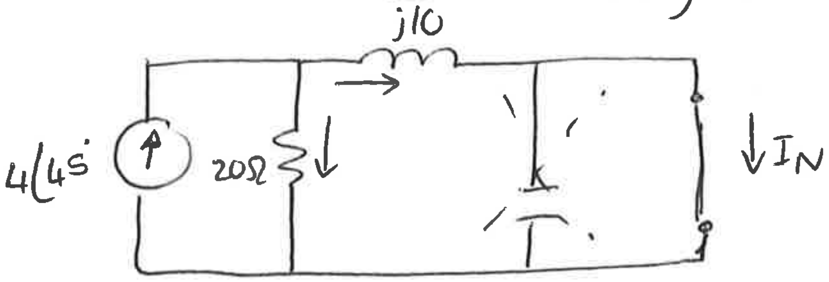
$= -j5 \parallel (20 + j10)$

$= 5{,}423 \angle -77{,}47^\circ$

↓ top.

(*) $V_{TH}$ or $I_N$

(transform ⊕ ⟶ ⊙)

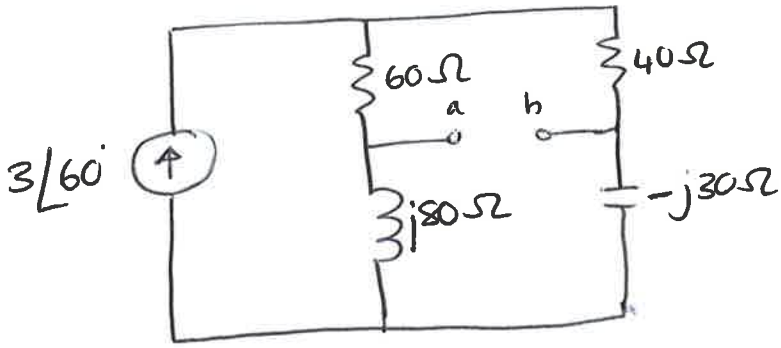



$$\therefore \quad I_N = \left(\frac{20}{20 + j10}\right) 4\angle 45^\circ$$

$$= (3{,}578\angle 18{,}43^\circ)\ A$$

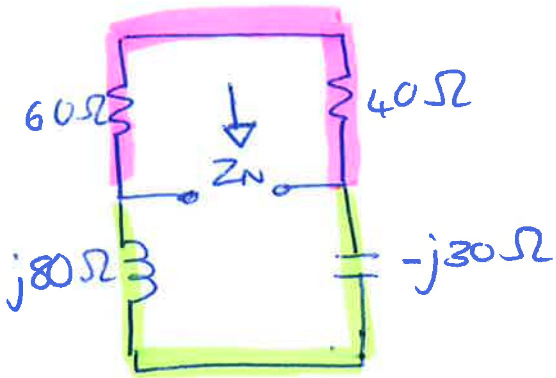
$$\therefore \quad V_{TH} = I_N Z_N = (19{,}4\angle -59^\circ)\ V.$$

10.64




Find the NORTON-equivalent at terminals a-b.

$\boxed{Z_N}$




$\therefore Z_N = (60+40) \,//\, (j80 - j30)$

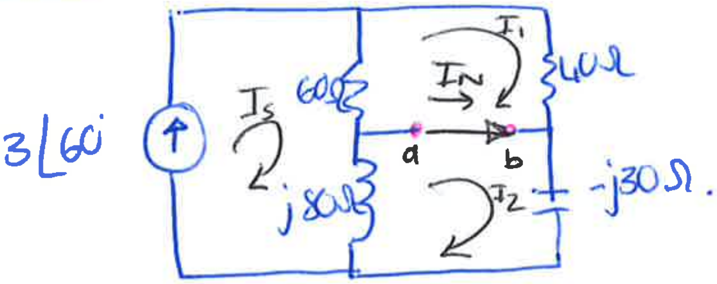
$= 100 \,//\, j50$

$= 20 + j40$

$= 44.72 \angle 63.43°$

$* \; 100 \,//\, j50 = \dfrac{(100)(j50)}{100 + j50}$

$\boxed{I_N}$




- $I_s = 3 \angle 60°$
- Mesh 1:

  $100 I_1 - 60 I_s = 0$

  $I_1 = 1.8 \angle 60°$
- Mesh 2:

  $(j80 - j30) I_2 - j80 I_s = 0$

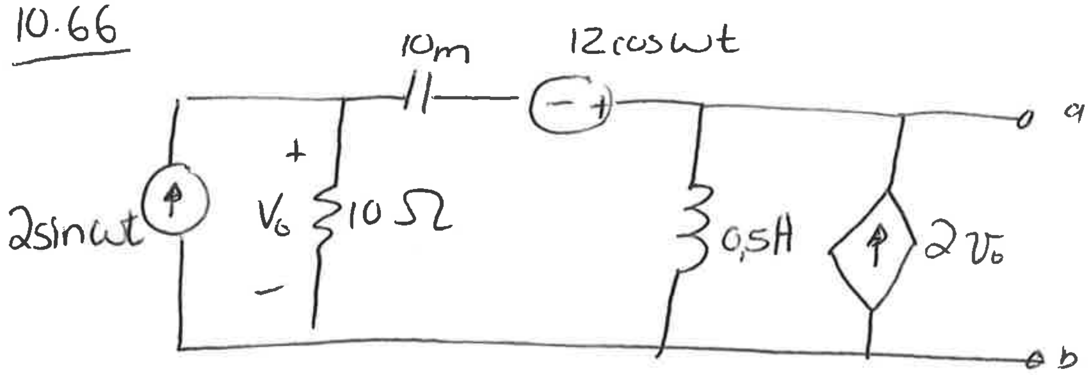
  $I_2 = 4.8 \angle 60°$

$\therefore I_N = I_2 - I_1$

$= 3 \angle 60°$

10.66

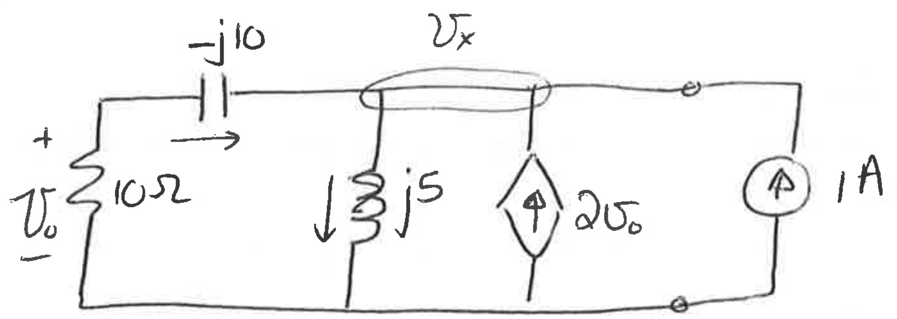





* Thevenin & Norton

* $Z_{TH} = Z_N$ (Dependant source ⇒ TEST SOURCE)

($\omega$ = 10 rad/s)




* $v_o = v_x\left(\frac{10}{10-j10}\right)$ → voltage division.

* @ node x:

$$\frac{v_x}{10-j10} + \frac{v_x}{j5} = 2v_o + 1$$

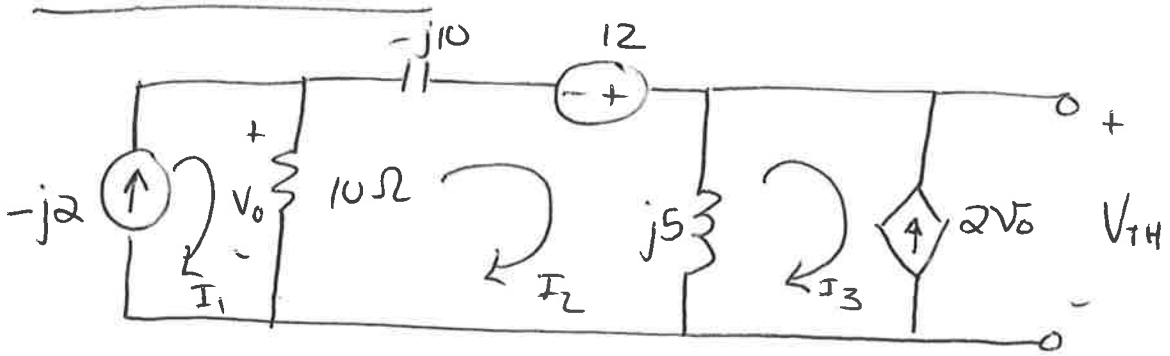
$$\frac{v_x}{10-j10} + \frac{v_x}{j5} - 2\left(\frac{10}{10-j10}\right)v_x = 1$$

$$\therefore v_x = \frac{-10+j10}{21+j2}$$

$$\therefore Z_N = Z_{TH} = \frac{V_x}{1}$$

$$= (0{,}670 \angle 129{,}56^\circ)\ \Omega$$

## $V_{TH}$ and $I_N$




* $I_1 = -j2$
* $I_3 = 2V_0$
* Loop 2:

$$\cancel{-10(-j2)}^{V_0} + (-j10)I_2 - 12 + j5I_2 - j5I_3 = 0$$

* $\boxed{V_0 = 10(I_1 - I_2)}$

* $j5(2V_0)$

$$-10(-j2) + 10I_2 - j10I_2 + j5I_2 - j10[-20j - 10I_2] = 12.$$

$$(10 - j105)I_2 = 12 - j20 - 200$$

$$I_2 = 0{,}01977 - j1{,}7924$$

$\therefore V_{TH} = j5(I_2 - I_3)$

$= (29{,}78\angle -3{,}615^\circ)\,V$

$V_0 = 10(-j2 - 0{,}0197 + j1{,}792)$

$= -0{,}1977 - j2{,}0764$

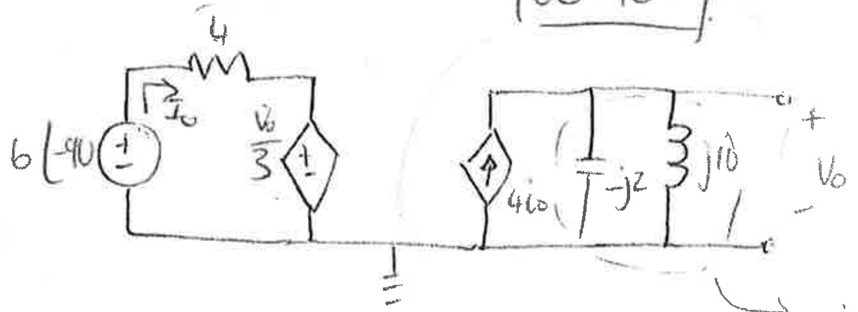
$\therefore I_N = \frac{V_{TH}}{Z_{TH}}$

$= (44{,}46\angle -133{,}16^\circ)\,A.$

10.68

$\boxed{\omega = 10}$

Thevenin




$-j2 // j10 = -j2{,}5$

$V_{TH}$ :

- $V_o = V_{TH} = 4I_o(-j2{,}5) = -j10 I_o$ —— ①
- $-6\angle{-90} + 4I_o + \frac{V_o}{3} = 0$ —— ②

① in ② $j6 + 4I_o + \frac{-j10 I_o}{3} = 0$

$(4 - j3{,}33) I_o = -j6$

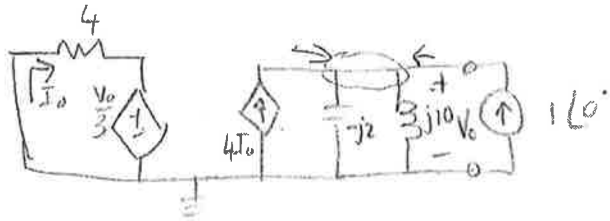
$I_o = 0{,}738 - j0{,}885$

$\therefore V_{TH} = -j10 I_o$

$= 11{,}52 \angle{-140{,}197^\circ}$

$\therefore \underline{V_{TH} = 11{,}52 \sin(10t - 50{,}197^\circ)}$

$Z_{TH}$ :




- $4I_o + \frac{V_o}{3} = 0$

$I_o = -\frac{V_o}{12}$ —— ①

- $1 + 4I_o = \frac{V_o}{-j2} + \frac{V_o}{j10}$ —— ②

① in ② $\Rightarrow V_o = 1{,}2293 - j1{,}4766$

$\therefore \underline{Z_{TH} = \frac{V_o}{1} = (1{,}2293 - j1{,}4766)\,\Omega}$